Neuroanatomy & Physiology: Autonomic Nervous System

1. The preganglionic neurons of the sympathetic nervous system are anatomically located in which specific region of the spinal cord?
a. Anterior horn of all spinal segments
b. Posterior horn of cervical segments
c. Dorsal root ganglia of lumbar segments
d. Mantle layer of the brainstem
e. Intermediolateral gray column from T1 to L2/L3
2. Because of its specific anatomical origin, the sympathetic division of the autonomic nervous system is formally designated as the:
a. Craniosacral outflow
b. Cervical outflow
c. Sacrococcygeal outflow
d. Thoracolumbar outflow
e. Peripheral outflow
3. The parasympathetic nervous system has preganglionic fibers arising from both the brainstem and the spinal cord, a pattern known as the:
a. Thoracolumbar outflow
b. Lumbosacral outflow
c. Craniosacral outflow
d. Cervicothoracic outflow
e. Sacrococcygeal outflow
4. Which of the following cranial nerves carries preganglionic parasympathetic fibers that arise from the Edinger-Westphal nucleus?
a. Oculomotor nerve (CN III)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X)
e. Trigeminal nerve (CN V)
5. Preganglionic sympathetic axons exit the anterior rami of spinal nerves to enter the paravertebral sympathetic chain ganglia via the:
a. Gray rami communicantes
b. Dorsal roots
c. White rami communicantes
d. Greater splanchnic nerves
e. Intercostal nerves
6. After entering the sympathetic trunk, postganglionic sympathetic fibers heading to the skin of the limbs and body wall rejoin the spinal nerves via the:
a. White rami communicantes
b. Anterior roots
c. Meningeal rami
d. Splanchnic rami
e. Gray rami communicantes
7. Which pair of structures represents the specialized prevertebral (collateral) sympathetic ganglia located anterior to the abdominal aorta?

a. Ciliary and pterygopalatine ganglia
b. Otic and submandibular ganglia
c. Superior cervical and stellate ganglia
d. Celiac and superior mesenteric ganglia
e. Chain ganglia and terminal ganglia

8. The sympathetic preganglionic fibers that form the greater splanchnic nerve pass through the chain ganglia without synapsing and terminate directly within the:
a. Superior cervical ganglion
b. Celiac ganglion
c. Submandibular ganglion
d. Otic ganglion
e. Inferior mesenteric ganglion

9. Which organ is entirely unique because its secretory cells receive direct innervation from preganglionic sympathetic fibers, functioning essentially as a modified sympathetic ganglion?
a. Thyroid gland
b. Pancreas
c. Submandibular gland
d. Parotid gland
e. Adrenal medulla

10. The parasympathetic preganglionic neurons that travel within the vagus nerve (CN X) originate primarily from which brainstem nucleus?
a. Nucleus ambiguus
b. Superior salivatory nucleus
c. Inferior salivatory nucleus
d. Dorsal motor nucleus of the vagus nerve
e. Nucleus solitarius

11. The parasympathetic sacral outflow arises from the S2 to S4 segments of the spinal cord. The preganglionic fibers emerge from the cord as:
a. White rami communicantes
b. Gray rami communicantes
c. Pelvic splanchnic nerves
d. Greater splanchnic nerves
e. Lumbar splanchnic nerves

12. In the autonomic reflex arcs, what neurotransmitter is universally secreted by all sympathetic and parasympathetic preganglionic neurons?
a. Norepinephrine
b. Epinephrine
c. Dopamine
d. Acetylcholine
e. Serotonin

13. The receptors located on the cell bodies of all postganglionic autonomic neurons within both sympathetic and parasympathetic ganglia are:
a. Muscarinic cholinergic receptors
b. Alpha-1 adrenergic receptors
c. Beta-1 adrenergic receptors
d. Nicotinic cholinergic receptors
e. Alpha-2 adrenergic receptors

14. Most postganglionic sympathetic nerve terminals release which neurotransmitter at their target visceral effector organs?
a. Acetylcholine
b. Norepinephrine
c. Dopamine
d. Serotonin
e. Epinephrine
15. Postganglionic sympathetic innervation to most thermoregulatory sweat glands is an exception to the general rule because these terminals secrete:
a. Norepinephrine
b. Epinephrine
c. Acetylcholine
d. Dopamine
e. Gamma-aminobutyric acid
16. All postganglionic parasympathetic nerve terminals release acetylcholine. The receptors found on the target visceral effector organs are:
a. Nicotinic receptors
b. Alpha receptors
c. Beta receptors
d. Muscarinic receptors
e. Dopaminergic receptors
17. Activation of the sympathetic nervous system causes pupillary dilation (mydriasis). This effect is mediated by the contraction of the pupillary dilator muscle via:
a. Muscarinic receptors
b. Nicotinic receptors
c. Alpha-1 adrenergic receptors
d. Beta-2 adrenergic receptors
e. Alpha-2 adrenergic receptors
18. Parasympathetic stimulation causes pupillary constriction (miosis) and lens accommodation for near vision. These actions are executed by the contraction of the sphincter pupillae and ciliary muscles via:
a. Alpha-1 receptors
b. Beta-1 receptors
c. Muscarinic receptors
d. Nicotinic receptors
e. Beta-2 receptors
19. Sympathetic stimulation increases the heart rate and myocardial contractility. This positive chronotropic and inotropic effect is mediated primarily by:
a. Alpha-1 receptors
b. Alpha-2 receptors
c. Beta-1 adrenergic receptors
d. Beta-2 adrenergic receptors
e. Muscarinic receptors
20. Bronchodilation (relaxation of bronchial smooth muscle) is triggered during a sympathetic fight-or-flight response through the activation of:
a. Beta-2 adrenergic receptors
b. Beta-1 adrenergic receptors

c. Alpha-1 receptors
d. Muscarinic receptors
e. Nicotinic receptors

21. Parasympathetic activation slows the heart rate down toward its baseline rhythm. This negative chronotropic effect is mediated at the sinoatrial node by:

a. Nicotinic receptors
b. M1 muscarinic receptors
c. M2 muscarinic receptors
d. Alpha-1 receptors
e. Beta-1 receptors

22. Visceral blood vessels supplying the skin and abdominal organs undergo vasoconstriction when sympathetic tone rises. This response is mediated by:

a. Beta-2 receptors
b. Beta-1 receptors
c. Muscarinic receptors
d. Alpha-1 adrenergic receptors
e. Alpha-2 receptors

23. The enteric nervous system handles the intrinsic regulation of the GI tract. The autonomic plexus located within the muscular wall layer responsible for controlling GI motility is the:

a. Submucosal plexus (Meissner's)
b. Myenteric plexus (Auerbach's)
c. Celiac plexus
d. Hypogastric plexus
e. Pelvic plexus

24. The baroreceptor reflex controls acute shifts in blood pressure. Afferent signals from the carotid sinus baroreceptors travel up to the brainstem via the:

a. Vagus nerve (CN X)
b. Trigeminal nerve (CN V)
c. Glossopharyngeal nerve (CN IX)
d. Sympathetic chain
e. Oculomotor nerve (CN III)

25. When blood pressure spikes, the baroreceptor reflex responds by driving a compensatory change. This involves an immediate increase in:

a. Sympathetic tone to the heart and blood vessels
b. Parasympathetic tone via the vagus nerve to slow the heart rate
c. Systemic epinephrine levels from the adrenal medulla
d. Renin production by the kidneys
e. Intrathecal cerebrospinal fluid pressure

26. The highest subcortical center responsible for coordinating and integrating autonomic, endocrine, and somatic homeostatic behaviors is the:

a. Thalamus
b. Hypothalamus
c. Cerebellum
d. Basal ganglia
e. Hippocampus

27. Postganglionic sympathetic fibers arising from the superior cervical ganglion travel through the head by forming networks around the:

a. Internal jugular vein
b. Internal and external carotid arteries
c. Vestibulocochlear nerve
d. Maxillary minor veins
e. Dural venous sinuses

28. Parasympathetic stimulation of the gastrointestinal tract leads to a coordinated visceral response characterized by:
a. Decreased motility and sphincter contraction
b. Complete arrest of glandular secretions
c. Increased motility, relaxation of sphincters, and increased secretions
d. Vasoconstriction of mesenteric blood vessels
e. Glycogenolysis in the liver parenchyma

29. The urinary bladder wall muscle (detrusor) and its internal sphincter are under reciprocal autonomic control. Micturition (urination) is driven by:
a. Sympathetic contraction of the detrusor muscle
b. Parasympathetic contraction of the detrusor and relaxation of the internal sphincter
c. Sympathetic contraction of the internal urethral sphincter
d. Voluntary skeletal relaxation of the detrusor muscle
e. Somatic pudendal activation of the gray rami

30. Which autonomic division is responsible for localized metabolic processes including glycogenolysis and lipolysis to mobilize energy substrates during stress?
a. Parasympathetic division
b. Enteric nervous system
c. Somatopleuric division
d. Sympathetic division
e. Craniosacral outflow

31. A 24-year-old male presents with a drooping upper eyelid (ptosis), a constricted pupil (miosis), and a lack of sweating (anhidrosis) on the left side of his face. This presentation describes:
a. Bell's palsy
b. Frey's syndrome
c. Raynaud's disease
d. Horner's syndrome
e. Hirschsprung disease

32. The clinical manifestation of Horner’s syndrome is directly caused by an interruption of which neurological pathway?
a. Postganglionic parasympathetic fibers from the ciliary ganglion
b. Cranial parasympathetic outflow from the oculomotor nerve
c. Cervical sympathetic pathway (oculosympathetic fibers)
d. Mandibular division sensory branches
e. Pelvic splanchnic nerve inputs

33. An infant fails to pass meconium within 48 hours of birth and develops progressive abdominal distension. Biopsy confirms an absence of myenteric and submucosal plexuses in the distal colon wall. This is:
a. Megaesophagus
b. Achalasia
c. Hirschsprung disease (Congenital megacolon)
d. Crohn's disease

e. Intussusception

34. Hirschsprung disease stems from a failure of which embryonic cell population to migrate normally into the wall of the gastrointestinal tract during the fifth week?

a. Splanchnic mesoderm cells
b. Endodermal lining cells
c. Paraxial somite neuroblasts
d. Neural crest cells
e. Surface ectoderm sheets

35. A 45-year-old female experiences painful blanching, cyanosis, and throbbing of her fingers when exposed to cold temperatures due to severe, episodic vasospasm of the digital arteries. This condition is:

a. Horner's syndrome
b. Raynaud's disease
c. Frey's syndrome
d. Guillain-Barré syndrome
e. Lambert-Eaton syndrome

36. A patient presents with localized facial flushing and sweating on their cheek exclusively while chewing food (gustatory sweating) following a parotidectomy. This pathology is:

a. Bell's palsy
b. Horner's syndrome
c. Trigeminal neuralgia
d. Frey's syndrome
e. Ramsay Hunt syndrome

37. Frey's syndrome occurs due to the misdirected, aberrant regeneration of parasympathetic secretomotor fibers traveling from which nerve into the sympathetic sweat glands of the skin?

a. Auriculotemporal nerve (carrying fibers from CN IX)
b. Chorda tympani nerve (carrying fibers from CN VII)
c. Greater petrosal nerve
d. Mental nerve
e. Lingual nerve trunk

38. A patient is treated with Atropine, a competitive muscarinic receptor antagonist, to reverse severe bradycardia. Which of the following side effects is expected due to the systemic block of parasympathetic actions?

a. Excessive salivation (sialorrhea)
b. Extreme pupillary constriction (miosis)
c. Xerostomia (dry mouth) and blurred vision (cycloplegia)
d. Diarrhea and abdominal cramping
e. Increased urinary frequency

39. A 60-year-old male with a history of a Pancoast tumor (apical lung carcinoma) develops new-onset Horner's syndrome. The tumor has directly invaded which autonomic structure?

a. Ciliary ganglion
b. Pterygopalatine ganglion
c. Sympathetic trunk / Stellate ganglion
d. Otic ganglion
e. Pelvic splanchnic nerves

40. Autonomic dysreflexia is a life-threatening clinical emergency that can develop in patients with spinal cord injuries at or above the T6 level. It is characterized by an unmodulated, massive explosive discharge from the:
a. Cranial parasympathetic system
b. Sacral parasympathetic system
c. Sympathetic nervous system below the level of the injury
d. Enteric nervous system
e. Somatic motor cortex
41. A patient with asthma is prescribed Albuterol, a selective beta-2 adrenergic receptor agonist. This drug successfully induces bronchodilation by mimicking the endogenous action of:
a. Acetylcholine on muscarinic receptors
b. Norepinephrine on alpha-1 receptors
c. Norepinephrine/Epinephrine on beta-2 receptors
d. Acetylcholine on nicotinic receptors
e. Serotonin on smooth muscle walls
42. A patient accidentally ingests an organic insecticide containing organophosphates, which permanently inhibit acetylcholinesterase. Which clinical sign confirms an overactivation of the parasympathetic system?
a. Extreme pupillary dilation (mydriasis)
b. Profuse salivation, lacrimation, urination, and defecation (SLUDGE syndrome)
c. Tachycardia and hypertension
d. Acute urinary retention
e. Severe bronchodilation
43. A patient is placed on Propropanol, a non-selective beta-adrenergic blocker, to manage hypertension. This drug drops blood pressure and heart rate by directly antagonizing:
a. Alpha-1 adrenergic receptors
b. Muscarinic cholinergic receptors
c. Beta-1 and Beta-2 adrenergic receptors
d. Nicotinic cholinergic receptors
e. Alpha-2 adrenergic receptors
44. Orthostatic hypotension, characterized by a sudden drop in blood pressure upon standing up, occurs when there is a functional failure or delay in the:
a. Accommodation reflex arc
b. Baroreceptor reflex arc
c. Gag reflex loop
d. Corneal reflex loop
e. Pupillary light reflex arc
45. A patient with a history of chronic dry mouth and dry eyes is treated with Pilocarpine, a direct-acting muscarinic agonist. This drug successfully increases salivation by activating:
a. Adrenergic receptors on salivary duct cells
b. Muscarinic cholinergic receptors on salivary acinar cells
c. Nicotinic receptors in the submandibular ganglion
d. Alpha-1 receptors on vascular smooth muscle
e. Sensory endings of the lingual nerve

46. A patient undergoes an evaluation for localized anhidrosis. The sweat glands are functionally unique because they are structurally innervated by sympathetic fibers, but use which anatomical pairing?

a. Preganglionic sympathetic fibers releasing acetylcholine
b. Postganglionic sympathetic fibers releasing acetylcholine onto muscarinic receptors
c. Postganglionic sympathetic fibers releasing norepinephrine onto alpha-1 receptors
d. Postganglionic parasympathetic fibers releasing acetylcholine
e. Somatic motor fibers releasing acetylcholine onto nicotinic receptors

47. During a neurophysiology experiment, a spinal cord transection is performed at the C7 level. The isolated lower segments display a temporary loss of all reflex activity, a clinical state called:

a. Autonomic dysreflexia
b. Spinal shock
c. Horner's crisis
d. Raynaud's phenomenon
e. Enteric paralysis

48. A patient experiences an intraoperative drop in blood pressure. The anesthesiologist administers Phenylephrine, a selective alpha-1 adrenergic agonist, which restores pressure by inducing:

a. An increase in heart rate
b. Severe bronchodilation
c. Systemic vasoconstriction of peripheral blood vessels
d. An increase in the secretion of pancreatic enzymes
e. A block of nicotinic receptors in the chain ganglia

49. Neostigmine is administered to a patient to reverse a neuromuscular block. Because it increases acetylcholine concentrations universally, it can cause unwanted visceral parasympathetic side effects by overstimulating:

a. Nicotinic receptors
b. Alpha-1 receptors
c. Beta-1 receptors
d. Muscarinic receptors
e. Alpha-2 receptors

50. The complex physiological regulation of the male reproductive tract involves sequential coordination between both autonomic divisions. Erections are mediated by the parasympathetic system, whereas ejaculation is driven by the:

a. Somatic pudendal nerve exclusively
b. Sympathetic nervous system
c. Enteric nervous system
d. Central canal ependymal cells
e. Volitional motor cortex